Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If the equation $x^2 + bx + ca = 0$ and $x^2 + cx + ab = 0$ have a common root and $b \neq c$, then their other roots will satisfy the equation

(A) $x^2 + (b + c)x + bc = 0$

(B) $x^2 - ax + bc = 0$

(C) $x^2 + ax + bc = 0$

(D) none of these

Q.2

The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one common root. The other roots of first and second equation are integers in the ratio 4 : 3. Then common root is

(A) 1

(B) 4

(C) 3

(D) 2

Q.3

If the equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a common root different from -1, then $|b|$ is equal to:

(A) 2

(B) 3

(C) $\sqrt{3}$

(D) $\sqrt{2}$

Q.4

If equations $ax^2 + bx + c = 0, (a, b, c \in \mathbf{R}, a \neq 0)$ and $2x^2 + 3x + 4 = 0$ have a common root, then $a : b : c$ equals:

(A) $1 : 2 : 3$

(B) $2 : 3 : 4$

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(C) $4:3:2$

(D) $3:2:1$

Q.5

If α, β, γ are the roots of the equation $x^3 + qx + r = 0$, then the equation whose roots are $-\alpha^{-1}, -\beta^{-1}, -\gamma^{-1}$ is:

(A) $rx^3 + qx^2 - 1 = 0$

(B) $rx^3 - qx^2 - 1 = 0$

(C) $rx^3 + qx^2 + 1 = 0$

(D) $rx^3 - qx^2 + 1 = 0$

Q.6

The number of distinct real roots of the equation
$(x+3)^4 + (x+5)^4 = 16$ (1)
is

(A) 1

(B) 2

(C) 3

(D) 4

Q.7

Sum and product of all the roots of the equation
$\left(x^2 - x - 1\right)\left(x^2 - x - 2\right)\left(x^2 - x - 3\right) \ldots \left(x^2 - x - 2012\right) = 0$ is

(A) $2012, 2012!$

(B) $-2012, 2012!$

(C) $-2012, -2012!$

(D) $2012, -2012!$

Q.8

If the equation $a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x = 0, a_1 \neq 0, n \geq 2$ has a positive root $x = \alpha$, then the equation $n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \ldots + a_1 = 0$ has a positive root, which is

(A) greater than or equal to α

(B) equal to α

(C) greater than α

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(D) smaller than α

Q.9

Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to:

(A) 6

(B) -6

(C) 3

(D) -3

Q.10

If $x = 1 + i$ is a root of $x^3 - ix + 1 - i = 0$, then the quadratic equation whose roots are the remaining two roots of $x^3 - ix + 1 - i = 0$ is

(A) $x^2 + (1 + i)x + 1 + i = 0$

(B) $x^2 + (1 + i)x + i = 0$

(C) $x^2 + 2(1 + i)x - 2 = 0$

(D) none of these

Q.11

The number of polynomials of the form $x^3 + ax^2 + bx + c \; \forall a, b, c \in 1, 2, 3, \ldots\ldots, 10$ which are divisible by $x^2 + 1$ is equal to

(A) 5

(B) 10

(C) 20

(D) 100

Q.12

The values of k for which the equations
$x^2 - kx - 21 = 0$ and $x^2 - 3kx + 35 = 0$ will
have a common roots are:

(A) $k = \pm 4$

(B) $k = \pm 1$

(C) $k = \pm 3$

(D) $k = 0$

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Q.13

If both the roots of $k\left(6x^2+3\right)+rx+2x^2-1=0$ and $6k\left(2x^2+1\right)+px+4x^2-2=0$ are common, then $2r-p$ is equal to

(A) -1

(B) 0

(C) 1

(D) 2

Q.14

If $\mathrm{p}, \mathrm{q}, \mathrm{r}$ are non-zero real numbers, the two equation, $2a^2x^2-2abx+b^2=0$ and $p^2x^2+3pqx+q^2=0$ have :

(A) no common root

(B) one common root if $2\mathrm{a}^2+\mathrm{b}^2=\mathrm{p}^2+\mathrm{q}^2$

(C) two common roots if $3\mathrm{pq}=2\mathrm{ab}$

(D) two common roots if $3\mathrm{qb}=2$ ap

Q.15

If the quadratic equations $ax^2+2cx+b=0$ and $ax^2+2bx+c=0\ (b\neq c)$ have a common root, then $a+4b+4c$ is equal to

(A) -2

(B) -1

(C) 0

(D) 1

Answers & Solutions

Q.1 Answer:

$x^2+ax+bc=0$

Solution:

Given the quadratic equation is

$\mathrm{x}^2+\mathrm{bx}+\mathrm{ac}=0$ **....(1)**

$x^2+cx+ab=0$ **...(2)**

Let roots of (1) be a and β and roots of (2) be a.y.

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common root $= \mathrm{a}$

$\therefore a^2 + ba + ac = a^2 + ca + ab$

$(b-c)c = ab - ac = a(b-c)$

$\therefore \mathbf{a = a}$

NOW $a + \beta = -b \quad a\beta = ac$

$a + y = -c \quad ay = ab$

$\frac{a\beta}{ay} = \frac{ac}{ab} \Rightarrow \frac{b}{y} = \frac{c}{b}$

$\therefore \beta = c, y = b$ **...(3)**

since, a **is a common root** $\therefore f(a) = 0$

$\Rightarrow \mathrm{a}^2 + \mathrm{ab} + \mathrm{ac} = 0$

$a(a+b) + ac = 0$ **since** $a = a$

$a^2 + ab + ac = 0$

$\Rightarrow a(a+b+c) = 0 \Rightarrow a + b + c = 0$ **of** $b + c = -a$ **...(4)**

Equation having roots β **and** $_\gamma$ **is**

$x^2 - (\beta + y)x + \beta y = 0$

$x^2 - (c+b)x + bc = 0$ **from (3)**

$x^2 - (-a)x + bc = 0$ **from** (4)

$\mathrm{x}^2 + \mathrm{ax} + \mathrm{bc} = 0$ **is the equation**

Q.2 Answer:

2

Solution:

Let roots of $x^2 - 6x + a = 0$ **be** $\alpha, 4m$ **and that of** $x^2 - cx + 6 = 0$ **be** $\alpha, 3m$ **where** m **is an integer We have,**

$\alpha + 4m = 6, \alpha(4m) = a$

and $\alpha + 3m = c, \quad \alpha(3m) = 6$

$\Rightarrow \quad \alpha m = 2 \Rightarrow a = 8$

Also, $\alpha + \frac{8}{\alpha} = 6 \quad \Rightarrow \quad \alpha^2 - 6\alpha + 8 = 0$

$\Rightarrow \quad (\alpha - 2)(\alpha - 4) = 0 \Rightarrow \alpha = 2, 4$

As m **is an integer,** $\alpha \neq 4 (\because \alpha m = 2)$

$\therefore \quad \alpha = 2$

Q.3 Answer:

$\sqrt{3}$

Solution:

$x^2 + bx - 1 = 0$ **(1)**

and $x^2 - x + b = 0$ **(2)**

Subtracting this to get common root as $x = \frac{b+1}{b-1}$

Substituting the common root in equation (2)

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$\left(\frac{b+1}{b-1}\right)^2\left(\frac{b-1}{b-1}\right)+b-0$

$\left(\frac{b+1}{b-1}\right)^2+\left(\frac{b-1}{b-1}\right)+b=0$

$(b+1)^2+(b+1)(b-1)-b(b-1)^2=0$

$b^2+2b+1+b^2-1+b\left(b^2+1-2b\right)=0$

$b^3+3b=0$

$b=0,+\sqrt{3i}$

0 is not possible.

$b=\sqrt{3}$

Q.4 Answer:

$2:3:4$

Solution:

As roots of $2x^2+3x+4=0$ are non-real complex numbers, the given quadratic equations have both roots in common.

Thus, $\frac{a}{2}=\frac{b}{3}=\frac{c}{4} \quad \Rightarrow a:b:c=2:3:4$

Q.5 Answer:

$\mathrm{rx}^3-\mathrm{qx}^2-1=0$

Solution:

where $\mathrm{a}, \mathrm{b}, \mathrm{c}$ **are coefficients of** $\mathrm{x}^3, \mathrm{x}^2, \mathrm{x}$ **respectively and** d **is the constant**

So in equation $x^3+qx+r=0$

$\rightarrow \alpha+\beta+\gamma=0$

$\rightarrow a\beta+\beta\gamma+\gamma\alpha=\mathrm{q}$

$\rightarrow \alpha\beta\gamma=-\mathrm{r}$

Now equation whose roots are $-\alpha^{-1},-\beta^{-1},-\gamma-1$ **must have sum** $=\frac{-1}{\alpha}+\frac{-1}{\beta}+\frac{-1}{\gamma}=\frac{-(\alpha\beta+\beta\gamma+\gamma\alpha)}{a\beta\gamma}=\frac{-\mathrm{q}}{\mathrm{r}}$

sum of products of two roots at a time

$=\frac{-1}{\alpha}\times\frac{-1}{\beta}+\frac{-1}{\beta}\times\frac{-1}{\gamma}+\frac{-1}{\gamma}\times\frac{-1}{\alpha}=\frac{-(\alpha+\beta+\gamma)}{\alpha\beta\gamma}=0$

product of roots $=\frac{-1}{\alpha\beta\gamma}=\frac{1}{\mathrm{r}}$

New equation

$=\mathrm{x}^3-\frac{\mathrm{q}}{\mathrm{r}}\mathrm{x}^2-\frac{-1}{\mathrm{r}}=0\rightarrow \mathrm{rx}^3-\mathrm{qx}^2-1=0$

Therfore Answer is B

Q.6 Answer:

2

Solution:

Put $x+4=t$**, so that (1) becomes** $(t-1)^4+(t+1)^4=16$

$\Rightarrow \quad 2\left(t^4+6t^2+1\right)=16$

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$\Rightarrow \quad t^4 + 6t^2 - 7 = 0$

$\Rightarrow \quad t^2 = 1, -7 \quad \Rightarrow \quad t = \pm 1, \sqrt{7}i$

Thus, the equation (1) has two real roots.

Q.7 Answer:

$2012, 2012!$

Solution:

We have,

$\left(x^2 + x - 2\right)\left(x^2 + x - 3\right) = 12$

Let, $\left(x^2 + x - 2\right) = y$

Then,

$y(y - 1) = 12$

$y^2 - y = 12$

$y^2 - y - 12 = 0$

$y^2 - (4 - 3)y - 12 = 0$

$y^2 - 4y + 3y - 12 = 0$

$y(y - 4) + 3(y - 4) = 0$

$(\mathrm{y} - 4)(\mathrm{y} + 3) = 0$

If, $y - 4 = 0$ **then,** $y = 4$

If, $\mathrm{y} + 3 = 0$ **then,** $\mathrm{y} = -3$

Put the value of $y = \left(x^2 + x - 2\right)$

$\left(x^2 + x - 2\right) = 4$

$x^2 + x - 6 = 0$

$x^2 + (3 - 2)x - 6 = 0$

$\mathrm{x}^2 + 3\mathrm{x} - 2\mathrm{x} - 6 = 0$

$x(x + 3) - 2(x + 3) = 0$

$(\mathrm{x} + 3)(\mathrm{x} - 2) = 0$

$\mathrm{X} = 2, -3$

If, $y = -3$

$x^2 + x - 2 = -3$

$x^2 + x - 1 = 0$

Then, sum of non real values $= -1$

Hence, this is the answer.

Q.8 Answer:

smaller than α

Solution:

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x$. **Then,** $f(0) = 0$ **and** $f(\alpha) = 0$.

By the Rolle's theorem $\exists \beta \in (0, \alpha)$ **such that** $f'(\beta) = 0$.

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Thus, $na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \ldots + a_1 = 0$
has a positive root β **which is smaller than** α**.**

Q.9 Answer:

3

Solution:

$\alpha + \beta = 6, \alpha\beta = -2$
$a_{10} - 2a_8 = \alpha^{10} + \beta^{10} + \alpha\beta\left(\alpha^8 + \beta^8\right)$
$= (\alpha + \beta)\left(\alpha^9 + \beta^9\right) = 6a_9$
$\Rightarrow \frac{a_{10} - 2a_8}{2a_9} = 3$

Q.10 Answer:

$x^2 + (1+i)x + i = 0$

Solution:

Let other roots be β, γ **then** $(1+i) + \beta + \gamma = 0$ **and** $(1+i)\beta\gamma = -(1-i)$
$\Rightarrow \beta + \gamma = -(1+i), \beta\gamma = i$

Q.11 Answer:

10

Solution:

$fx = x^3 + ax^2 + bx + c$ **is divisible by** $x^2 + 1$ **or** $x + ix$ - i

So, $fi = 0$ **and** f-$i = 0$

$fi = 0 \Rightarrow i^3 + ai^2 + bi + c = 0$

$\Rightarrow$ - i - $a + bi + c = 0$

$\Rightarrow c$ - $a + ib$ - $1 = 0 \Rightarrow c = a,\ b = 1$

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$f\text{-}i = 0 \Rightarrow \text{-}i^3 + a\text{-}i^2 + b\text{-}i + c = 0$

$\Rightarrow i - a - ib + c = 0$

$\Rightarrow c - a - ib - 1 = 0 \Rightarrow c = a, \quad b = 1$

$b = 1$ **and** $c = a$

$\Rightarrow 10$ **polynomials are possible**

Q.12 Answer:

$k = \pm 4$

Solution:

Let α **be the common root to the equations:**

$x^2 - kx - 21 = 0$ **and** $x^2 - 3kx + 35 = 0$

$\therefore \quad \alpha'$ **satisfies both the equations**

$\therefore \quad \alpha^2 - k\alpha - 21 = 0$

. . . (i) and $\alpha^2 - 3k\alpha + 35 = 0$ **...(ii)**

From (i) and (ii), $\alpha^2 - 21 = \frac{\alpha^2 + 35}{3}$

$\Rightarrow 3\alpha^2 - 63 = \alpha^2 + 35$

$\Rightarrow \alpha^2 = 49 \Rightarrow \alpha = \pm 7$

Now, again by eliminating α^2 **from (i) and (ii), we get**

$\mathrm{k}\alpha + 21 = 3\mathrm{k}\alpha - 35$

$\Rightarrow 2k\alpha = 56 \Rightarrow k = \frac{56}{2\alpha}$

When $\alpha = 7$ **then** $k = 4$ **When** $\alpha = -7$ **then** $k = -4$

Hence, $k = \pm 4$

Q.13 Answer:

0

Solution:

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Given equation can be written as

$(6k+2)x^2+rx+3k-1=0$

and $2(6k+2)x^2+px+2(3k-1)=0$

Condition for common root is

$\frac{12k+4}{6k+2}=\frac{p}{r}=\frac{6k-2}{3k-1}=2$ **or** $2r-p=0$

Q.14 Answer:

no common root

Solution:

Discriminant of the equation

$2a^2x^2-2abx+b^2=0$ **is** $-4a^2b^2<0$ **and that of**

the equation

$\mathrm{p^2x^2+3pqx+q^2=0}$ **is** $\mathrm{5p^2q^2>0}$

There cannot be any common root.

Q.15 Answer:

0

Solution:

Let α **be the common root of the given equation. Then** $a\alpha^2+2c\alpha+b=0$ **and** $a\alpha^2+2b\alpha+c=0$

$\Rightarrow \quad 2\alpha(c-b)+(b-c)=0\Rightarrow\alpha=\frac{1}{2} \quad [\because b\neq c]$

Putting $\alpha=\frac{1}{2}$ **in** $a\alpha^2+2c\alpha+b=0$**, we get** $a+4b+4c=0$

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